

Title

Preoperative intranasal dexmedetomidine for prevention of acute encephalopathy after major joint reconstruction in older adults: study design and rationale

Abstract

Acute postoperative neurocognitive dysfunction represents a serious complication affecting morbidity and length of stay in elderly surgical patients undergoing joint replacement. Sleep disruption and neuroinflammation are postulated mechanisms. The alpha-2 adrenergic agonist dexmedetomidine exerts sedative, anxiolytic, and neuroprotective properties in laboratory models. We present the protocol for a double-blind, placebo-controlled, single-center trial evaluating preoperative intranasal dexmedetomidine for delirium prevention. A total of 240 patients aged 60 or older scheduled for total hip or knee arthroplasty will be randomized 1:1 to receive either dexmedetomidine (100 µg total) or saline via nasal instillation the evening before surgery. Primary endpoint is delirium incidence assessed by standardized cognitive screening within 72 hours post-op. Secondary endpoints include delirium duration, severity grading, nocturnal sleep architecture, pain intensity, and perioperative anxiety. Safety assessments include vital sign stability and adverse respiratory events.